

Runbook Template

Runbook Title	checkout-api Latency & Error-Rate Triage
Version	2.0 (see Section 10)
Last Updated	2026-07-11
Owner	checkout-platform
Reviewers	checkout-platform team lead; infra-platform on-call lead (added -- no formal review cycle existed in the source project)
Next Review Date	2026-10-11 (added -- quarterly review cadence assumed, not previously defined)

1. Overview

1.1 Purpose

Guide an on-call engineer through triaging elevated latency or error rate on checkout-api: identify which known failure pattern applies (or rule both out), apply an immediate mitigation where one exists, and know when to escalate rather than continue investigating locally.

1.2 Scope

Covers checkout-api only, triggered by two specific alerts. Explicitly does not cover: deploying, rolling back, or hotfixing checkout-api (handled entirely at the guardrail/orchestrator layer, see Section 5); API request/response schema details; deployment pipeline mechanics for any service.

1.3 Target Audience

checkout-api on-call SRE (L1 first responder).

1.4 Prerequisites

- Access to the checkout-api-overview Grafana dashboard (Postgres Pool / Redis / Downstream Latency / Errors-by-status-code panels)
- psql access to the PgBouncer admin console for pool-size changes
- Access to #inc-checkout-latency and #infra-oncall Slack channels

2. Triggers

2.1 When to Use This Runbook

Trigger Type	Description
Alert Name	checkout-api-p99-latency-high
Threshold	p99 > 1500ms for 5 min
Alert Name	checkout-api-error-rate-high
Threshold	5xx+4xx rate > 2%

Monitoring Dashboard	checkout-api-overview (Grafana)
-----------------------------	---------------------------------

2.2 Related SLOs

(Added -- not formally tracked as distinct SLOs in the source project prior to this template; targets below are set equal to the alert thresholds for now.)

SLO Name	Target	Current Status
Latency P99	< 400ms	See Grafana dashboard
Error Rate	< 0.1%	See Grafana dashboard

3. Diagnostic Steps

Step 1: Determine which alert fired

Objective: Route to the correct diagnostic path -- the two alerts lead to different investigations, not one shared checklist.

Finding	Next Step
p99-latency-high fired	Proceed to Step 2
error-rate-high fired, latency normal	Skip to Step 3

Step 2: Latency path

1. Check connection pool dashboard (active_connections vs max_connections) -- if pinned at 200, go to Section 4.1 row 1
2. Check cache hit ratio -- if dropped sharply (~95% -> under 50%), go to Section 4.1 row 2
3. Check downstream dependency latency panel -- if a dependency spiked at the same time, escalate to that dependency's on-call (Section 6), no local fix exists

Step 3: Error-rate path (latency normal)

1. Break down error rate by status code class (4xx vs 5xx)
2. 400: check for a recent deploy changing the request schema, or a client not yet updated
3. 401: check for a recent token-rotation or auth-service change
4. 5xx: check application error logs for stack traces; check correlation with a recent deploy

No documented mitigation exists yet for this path -- diagnostic steps only. If unresolved, escalate (Section 6).

4. Resolution Actions

4.1 Common Fixes

Issue	Resolution	Est. Time
Connection pool exhaustion (active_connections pinned)	Increase PgBouncer pool size (default_pool_size/max_client_conn), reload via admin console -- check headroom against max_connections (200) first	5-10 min

Cache node failover (cache_hit_ratio dropped)	Confirm cluster_state:ok via redis-cli; if not, page #infra-oncall. If ok, monitor -- self-resolves as cache repopulates	10-15 min
Error-rate path (4xx/5xx)	No documented fix -- diagnostic only, see Step 3	N/A

4.2 Service Restart Procedure

Not applicable. Neither known issue in this runbook is resolved by restarting a service. Connection-pool exhaustion is mitigated by a config-level pool-size increase, not a restart. Cache-node failover self-resolves via Redis Cluster's own automatic failover with no manual restart step.

5. Rollback Procedure

Deliberately not documented here. Rollback/deploy actions are never handled within runbook content in this project -- the agent may only draft a recommendation, and a human must explicitly approve and execute it, enforced at the guardrail/orchestrator layer independently of whether this runbook is retrieved correctly. Do not add rollback steps to this section.

6. Escalation Matrix

6.1 Escalation Criteria

Condition	Action
Unresolved after 15 min	Escalate to checkout-platform team lead, open #inc-checkout-latency
Unresolved after 30 min, or customer-facing revenue impact	Declare SEV1, page incident commander rotation

6.2 Escalation Contacts

Level	Team/Person	Contact Method	Response SLA
L1	checkout-api on-call	Follows this runbook	0-15 min
L2	checkout-platform team lead	#inc-checkout-latency	15-30 min
L3	Incident commander rotation	Page	30+ min

7. Communication Templates

(Added -- not previously drafted; included here since the template calls for it.)

7.1 Initial Incident Notification

Subject: [SEV] checkout-api - elevated latency/errors

Impact: Checkout requests slower or failing for a subset of users

Status: Investigating

Started: <time>

Next Update: <time + 15 min>

7.2 Resolution Notification

Subject: [RESOLVED] checkout-api - elevated latency/errors

Resolution: <what fixed it, e.g. PgBouncer pool size increased>

Duration: <total incident time>

Follow-up: <postmortem scheduled / not needed>

8. Post-Incident Actions

- Verify monitoring/alerting captured the issue (resolution criteria: p99 < 400ms for 10 min, error rate < 0.1%, no further alerts for 15 min)
- Update incident ticket with resolution details (added -- assumes a ticket system, not previously specified)
- Schedule postmortem if incident lasted > 20 minutes or had customer impact (added, consistent with existing postmortems in this corpus)

9. Reference Information

9.1 Related Documentation

Document	Link
Postmortem -- connection pool exhaustion	postmortems/2026-05-checkout-outage.md
Postmortem -- cache node failover	postmortems/2026-03-checkout-outage-cache.md
Code doc -- connection pooling	code-docs/checkout-api-connection-pooling.md
Code doc -- request schema	code-docs/checkout-api-request-schema.md (referenced by Step 3; not yet written)

9.2 Log Locations

Component	Log Path	Retention
checkout-api (synthetic)	synthetic-data/logs/checkout-api-*-app-logs.jsonl	N/A -- synthetic dataset

10. Version History

Version	Date	Changes
1.0	earlier in project	Initial draft. Framed connection-pool exhaustion as deploy-caused by default; mitigation was restart PgBouncer + cap HPA max replicas; included a rollback/deploy-refusal section directly inside the runbook.
2.0	2026-07-11	Corrected: root cause decoupled from any single presumed cause; mitigation replaced with PgBouncer pool-size increase; rollback/deploy content removed and moved to the guardrail layer.

This runbook was last tested on: not yet tested against a live incident -- synthetic corpus only.
Next scheduled test: not yet scheduled.